

ISSAC ROY

U.S. Citizen, San Diego, CA

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Education

University of California San Diego (3.98 GPA)

March 2027

B.S. Data Science | Math Minor | Cognitive Science Minor

San Diego, CA

Courses: Data Structures, Computer Vision, Systems Programming, Computer Organization, Database Management

Experience

Apple | *Software Engineering Intern, WebKit Developer Productivity*

June 2026 – Present

- Designing an automated telemetry system to detect, isolate, and quarantine non-deterministic “flaky” tests across WebKit, surfacing concurrency races and corrupted memory state hidden inside a massively distributed C++ test suite.
- Building the detection pipeline in C++ and Python, instrumenting test executions to capture per-run state, statistically classifying flakiness, and auto-quarantining offending tests to keep CI signal trustworthy.
- Resolving core operational bugs and optimizing the compiler pipeline across distributed build infrastructure, stabilizing CI/CD workflows that build one of the largest C++ codebases in the world for thousands of concurrent developers.

MC Computer Vision Lab at UCSD | *Research Assistant*

January 2026 – May 2026

- Reproduced an OpenRooms++ generative pipeline on NRP’s Kubernetes cluster, implementing methods from recent papers to chain Trellis2 and Flux 2.0 diffusion models that turn coarse meshes into high-fidelity 3D assets.
- Implemented inverse rendering algorithms from the literature (SAM3D, Laval Indoor HDR) to decompose generated imagery into PBR texture maps (Albedo, Roughness) for photorealistic reconstruction.

Neurocrine Biosciences | *Cloud Engineer Intern*

June 2025 – September 2025

- Saved \$200k in annual OpEx by architecting and deploying an end-to-end serverless, containerized data ingestion pipeline on AWS ECS and Lambda with Terraform (IaC), Docker, and multi-environment CI/CD.
- Built an AI-driven pipeline with AWS Bedrock, wiring LLM endpoints to AWS Redshift to convert raw patient data into study-specific narratives and save 50+ hours per month for study teams.

Projects

Turbo 3D Engine | *C++, SDL2, Linear Algebra*

March 2025

- Built a 3D graphics engine from scratch in C++, implementing a custom rasterization pipeline, matrix transformations for camera space, texture mapping, and lighting without external graphics libraries.
- Optimized rendering performance through manual memory management, efficient vertex processing algorithms, and a custom linear algebra library.
- Wrote an OBJ file parser to load and render 3D models from file, implementing a custom material system, texture mapping, and multiple lighting models for realistic rendering.

Project Daedalus | *C++, KiCad, Onshape, Arduino*

October 2025 – Present

- Developed a custom C++ flight computer on Arduino Nano Every, integrating MPU6050, BMP388, and LoRa radio for 915Hz telemetry and black-box logging.
- Designed a custom PCB in KiCad and a 3D-printed avionics bay in Onshape to sustain high-G loads and protect electronics from the ejection charge, targeting a 5,500ft (1 mile+) apogee.

Self-Playing Guitar | *Arduino, CAD, Laser Cutting, 3D Printing*

December 2023

- Won 1st place at IEEE’s Fall Competition against 30+ teams by engineering a self-playing guitar with custom laser-cut and 3D-printed parts, programming a note-transcriber algorithm to map musical input to precise servo motions.

Leadership

AWS Cloud Club at UCSD | *Founder & VP Technical*

November 2025 – Present

- Established UCSD’s official AWS affiliate organization, building a technical community focused on cloud computing, serverless architecture, and industry best practices.
- Leading technical workshops on core AWS services (S3, Lambda, SageMaker), equipping 200+ members with the practical skills to build and deploy scalable cloud infrastructure.

Data Science Student Society (DS3) | *Director of Software*

October 2024 – Present

- Led a 10-person team to build a centralized membership portal for 1,000+ users, establishing GitHub Actions CI/CD and a scalable Supabase/Postgres backend with secure authentication flows.

Technical Skills

Languages: C++, C, Python, Java, SQL, Bash, TypeScript, React

Tools: AWS, Linux, Terraform, Docker, Git, Kubernetes, Postgres, MongoDB, Arduino, PyTorch, OpenCV

Certifications: AWS Cloud Practitioner, AWS AI Practitioner, AWS Solutions Architect Associate